

Erdős Problem #425

EP-425

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $F(n)$ be the maximum possible size of a subset $A \subseteq \{1, \dots, N\}$ such that the products ab are distinct for all $a < b$. Is there a constant c such that $F(n) = \pi(n) + (c + o(1))n^{3/4}(\log n)^{-3/2}$? If $A \subseteq \{1, \dots, n\}$ is such that all products $a_1 \cdots a_r$ are distinct for $a_1 < \cdots < a_r$, then is it true that $|A| \leq \pi(n) + O(n^{\frac{r+1}{2r}})$?

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